

$$\beta \rightarrow aA/a$$

13 → 6 13 16

$C \rightarrow c$

78. Chomsky Normal Form

CMF: we have a restriction ~~on the~~:

element in RHS should either be two variables or a Terminal

$$\begin{cases} A \rightarrow a \\ A \rightarrow BC \end{cases}$$

Steps: CFG $\rightarrow$ CNF

Step 1:

If the Start Symbol S occurs on some right side, create a new start symbol S' and a new production $S' \rightarrow S$

Step 2:

Remove Null Production. (Using the Null production Removal discussed in previous Lecture)

Step 3:

Remove Unit Productions

Step 4: Replace each production on $A \rightarrow B_1, \dots, B_n$ where $n \geq 2$, with $A \rightarrow B_1, C$ where $C \rightarrow B_2 \dots B_n$

Repeat this step for all productions having two or more symbols on the right side.

Step 5: If the right side of any production is in the form $A \rightarrow aB$ where ' a ' is a terminal and A and B are non-terminals, then the production is replaced by $A \rightarrow XB$ and $X \rightarrow a$. Repeat



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this step for every Production which is of the form $A \rightarrow \alpha B$

(79) Example of CFG to CNF

P: $S \rightarrow ASA | \alpha B$, $A \rightarrow B | S$, $B \rightarrow b | \epsilon$

steps
① $S' \rightarrow S$, $S \rightarrow ASA | \alpha B$, $A \rightarrow B | S$, $B \rightarrow b | \epsilon$

② remove $B \rightarrow \epsilon$ & $A \rightarrow \epsilon$

① p: $S' \rightarrow S$, $S \rightarrow ASA | \alpha B | \alpha$, $A \rightarrow B | S | \epsilon$, $B \rightarrow b$

② p: $S' \rightarrow S$, $S \rightarrow ASA | \alpha B | \alpha | SA | AS | S$

$A \rightarrow B | S$

$B \rightarrow b$

3) remove Unit Production $S \rightarrow S$, $S' \rightarrow S$, $A \rightarrow B$ and $A \rightarrow S$

① remove $S \rightarrow S$:

$S' \rightarrow S$, $S \rightarrow ASA | \alpha | SA | AS$, $A \rightarrow B | S$, $B \rightarrow b$

② remove $S' \rightarrow S$:

$S' \rightarrow ASA | \alpha | SA | AS$, $S \rightarrow ASA | \alpha | SA | AS$, $A \rightarrow B | S$, $B \rightarrow b$

③ remove $A \rightarrow B$

$\cup$; $A \rightarrow b | S$, $B \rightarrow b$

④ remove $B \rightarrow S$

$\cup$, $A \rightarrow b | ASA | \alpha | SA | AS$, $B \rightarrow b$



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~~4~~

After $\Rightarrow$ ~~A~~ $P: S' \rightarrow ASA/aB/a/AS/SA$

$S \rightarrow ASA/aB/a/AS/SA$

$A \rightarrow b/ASA/aB/a/AS/SA$

$B \rightarrow b$

4) find more than two variables in R/Hs

~~A~~ $S' \rightarrow ASA$ $S \rightarrow ASA$ and $A \rightarrow ASD$

$\Rightarrow P: S' \rightarrow AX/aB/a/AS/SA$

$S \rightarrow AX/aB/a/AS/SA$

$A \rightarrow b/AX/aB/a/AS/SA$

$B \rightarrow b$

$X \rightarrow SA$

5) Now change the productions $S' \rightarrow aB$, $S \rightarrow aB$

$A \rightarrow aB$

set $Y \rightarrow a$ and replace with a



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- ① if S in right, add $s' \rightarrow S$
- ② remove null production
- ③ remove Unit production
- ④ replace two more variables production in right hand with new production
- ⑤ replace other ~~the~~ illegal production " $\alpha\beta$ "

⑩ Greibach Normal Form

A CFG is Greibach Normal Form if :

$$\begin{cases} A \rightarrow b \\ A \rightarrow b C_1 C_2 \dots C_n \end{cases} \quad \text{where } A, C_1, C_2, \dots$$

is Non-Terminal Symbol

Steps:

- 1) Remove Unit Productions and Null Productions
- 2) check whether CFG has in chomsky Normal Form
---> it to CNF if it is not